1. Competency #2: Judge whether a mathematical result makes sense in the context of the problem

> *Identify whether a mathematical result or conclusion does, or does not, make sense given the context of a problem*

Highly relevant.
Students must understand that "viewed from the top" means **only the top face of each visible die counts**—not side or bottom faces. Including extra faces or assuming all seven dice are visible would yield an unreasonable answer. Additionally, if the computed sum falls outside the feasible range (e.g., less than 5 or greater than 30 for five dice), students should recognize the error. This reflects a strong grasp of **contextual constraints** (line of sight, physical structure) and the ability to validate results against real-world plausibility.

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2. Competency #9: Understand the relationship between problem context and mathematical representation, and use this to interpret the feasibility and limitations of solutions

> *Understand the relationship between the context of the problem and representation of the mathematical solution...*

Highly relevant.
The problem translates a **3D physical object** (stacked dice) into a **2D top-down diagram**, which is then abstracted into a **numerical calculation** (sum of pips). Students must correctly map between "real object → visual representation → mathematical model (numbers + rules)." For instance, seeing a side face showing "3" allows inference about possible top-face values using the "opposite faces sum to 7" rule. This bidirectional translation between reality and mathematical formalism is precisely what Competency #9 emphasizes.

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3. Competency #1: Interpret a mathematical result back into the real-world context

> *Interpret a mathematical result back into the real world context*

Moderately relevant.
Although the final answer is a number (total pips), it represents the **actual count of visible dots from above**. Students should recognize that this number isn't abstract—it corresponds to a real visual observation. For example, a sum of 23 implies a plausible combination like top faces showing 6, 6, 5, 4, and 2. This demonstrates the ability to reconnect a numerical result to its tangible meaning.

4. Competency #6: Use real-world knowledge to judge how to adapt or apply mathematical results

> *Use knowledge of how the real world impacts the outcomes... to make contextual judgments*

Moderately relevant.
The critical piece of "real-world knowledge" here is the **convention that opposite faces of a standard die sum to 7**. This is not a mathematical theorem but a **manufacturing standard**. Students must incorporate this external fact to reason about hidden faces and deduce visible ones. Ignoring or misremembering this rule would invalidate the entire solution, highlighting the necessity of integrating real-world conventions into mathematical reasoning.

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5. Competency #5: Interpret mathematical results presented in various formats and compare their relationship to the context

> *Interpret mathematical outcomes in a variety of formats... compare representations*

Low relevance.
The problem primarily involves a single representational shift: **diagram (top view) → numerical sum**. It doesn't require comparing multiple representations (e.g., tables, equations, 3D models), so this competency plays a minor role.

6. Competencies #3 & #4: Identify model limitations / Use tools to verify reasonableness

> *Identify the limits of the model... Use tools to ascertain reasonableness*

Low relevance.
The model is simple (dice rules + occlusion), with no complex assumptions. The problem doesn't involve computational tools or require evaluating modeling limitations (e.g., ignoring microscopic gaps between dice), which are typically beyond the scope of such contests.

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7. Competencies #7 & #8: Construct explanatory arguments / Recognize the scope and limits of mathematical concepts

> *Construct explanations... Recognise extent and limits of mathematical concepts*

Indirectly relevant.
Strong students might explain *why* a particular die must show a certain number on top, involving logical justification. However, the problem only asks for a numerical answer, not a written explanation. Competency #8 is minimally engaged here.

1. 第2条：判断数学结果在问题情境中是否合理

> *Identify whether a mathematical result or conclusion does, or does not, make sense given the context of a problem*

高度相关。
学生必须理解“从顶部看”意味着**只能看到每个可见骰子的顶面**，不能把侧面或底面算进去。如果有人把所有外露面都加起来，或误以为能看到7个骰子，结果就会荒谬。此外，若算出的点数不在合理范围（如5个骰子顶面点数总和小于5或大于30），也应能自我察觉错误。这体现了对**情境约束**（视角限制、骰子物理结构）的理解与结果合理性检验。

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2. 第9条：理解问题情境与数学表示之间的关系，并据此解释解的可行性与局限

> *Understand the relationship between the context of the problem and representation of the mathematical solution...*

高度相关。
本题将**三维物理结构**（骰子堆叠）转化为**二维俯视图**，再进一步抽象为**数字计算**（点数求和）。学生需在“实物 → 视图 → 数学模型（点数+规则）”之间建立正确映射。例如：看到某个骰子侧面是3点，就可推知其顶面不可能是4（因为3的对面是4，若3在侧面，则4可能在顶/底/另一侧，需结合方向判断——但核心是利用“对面和为7”的模型）。这种从现实对象到数学符号的转换及其反向解释，正是第9条的核心。

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3. 第1条：将数学结果还原到现实情境中

> *Interpret a mathematical result back into the real world context*

中等相关。
虽然本题最终答案是一个数字（总点数），但这个数字代表的是“从上方实际能看到的黑点数量”。学生需意识到这个数字不是抽象和，而是**对应真实视觉观察的结果**。例如，若计算得23点，应能想象这是5个骰子顶面分别显示如6、6、5、4、2等组合。这体现了从计算回到现实意义的过程。

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4. 第6条：利用现实世界知识判断如何调整或应用数学结果

> *Use knowledge of how the real world impacts the outcomes... to make contextual judgments*

中等相关。
这里的关键“现实知识”是：**标准骰子的对面点数之和为7**。这不是纯数学公理，而是**现实制造规范**。学生必须调用这一外部常识来推理不可见的面，从而确定可见面的可能值。若忽略此规则（或误记为其他和），整个模型就错了。因此，该题依赖于对“现实物体属性”的准确纳入数学推理。

5. 第5条：解读多种格式的数学结果，并比较其与情境的关系

> *Interpret mathematical outcomes in a variety of formats... compare representations*

低相关。
本题主要涉及**单一表示形式**：俯视图（图形）→ 数字和。虽有图与数的转换，但未要求比较不同表示（如表格、公式、3D模型等），故相关性较弱。

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6. 第3条 & 第4条：识别模型的局限性 / 使用工具验证解的合理性

> *Identify the limits of the model... Use tools to ascertain reasonableness*

低相关。
本题模型简单（骰子规则+视线遮挡），无复杂假设，也不涉及使用计算机模拟或工具验证。模型局限性（如忽略骰子微小缝隙、假设完美堆叠）在小学/初中竞赛中通常不考虑。

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7. 第7条 & 第8条：构建解释论证 / 认识数学概念的限度

> *Construct explanations... Recognise extent and limits of mathematical concepts*

间接相关。
优秀学生可能会解释“为什么这个骰子顶面一定是5”，这涉及论证。但题目本身只要求答案，未明确要求说理。第8条在此题中体现不明显。